Unit: Unit 0: Foundations of Algebra **Name:** ____________

Topic: GCF LCM **Date:** __________

Challenge (Habanero)

Q1. A recipe for $3 \cdot 4$ pancakes requires $2 \cdot 3$ eggs. What is the GCF of the number of pancakes and eggs?
(A) 2
(B) 3
(C) 6
(D) 12
(E) 1

Q2. A construction site has 12 and 15 meter long wooden planks. What is the LCM of these lengths?
(A) 30
(B) 60
(C) 120
(D) 180
(E) 240

Q3. A chef needs to package 48 cookies into boxes that hold 8 or 12 cookies. What is the GCF of the number of cookies per box?
(A) 2
(B) 4
(C) 8
(D) 12
(E) 1

Q4. A builder is constructing a fence with 6 and 9 foot long sections. What is the LCM of these lengths?
(A) 18
(B) 27
(C) 36
(D) 54
(E) 108

Q5. A recipe for $2 \cdot 3$ cakes requires $5 \cdot 2$ cups of flour. What is the LCM of the number of cakes and cups of flour?
(A) 30
(B) 60
(C) 120
(D) 180
(E) 240

Q6. A construction crew needs to mix 20 and 30 liters of cement. What is the GCF of the amounts of cement?
(A) 2
(B) 5
(C) 10
(D) 20
(E) 1

Q7. A chef is preparing a dish that requires 9 and 12 cups of ingredients. What is the LCM of the number of cups?
(A) 36
(B) 54
(C) 72
(D) 108
(E) 144

Q8. A builder is designing a building with 8 and 12 foot high walls. What is the GCF of the wall heights?
(A) 2
(B) 4
(C) 8
(D) 12
(E) 1

Open Ended Questions (FRQ)

Q9. A construction site has 24 and 30 meter long steel beams. Find the GCF and LCM of these lengths and explain how they can be used to design a stable structure.

Q10. A chef is preparing a recipe that requires $2 \cdot 3$ and $5 \cdot 2$ cups of ingredients. Find the LCM of the number of cups and explain how it can be used to scale up the recipe.

Chef's Tips

- Consider using visual aids to help students understand GCF and LCM concepts
- Encourage students to use prime factorization to find GCF and LCM
- Remind students to check their work by verifying that the GCF divides both numbers and the LCM is divisible by both numbers